to potential-free N/O or N/C contacts with a +5V rail. Energise the 5V regulated power supply to the circuit and connect the micro USB port of the STM32 board to a laptop/desktop PC, with the Arduino source code 'fault_finder_STM32_8IO.ino' open and the serial monitor window active. Select the serial monitor baud rate to 200000, as specified in the program.

Now change the logic state of one or more input pins by shorting or opening the external contacts and observe the following printed on the serial monitor window on the PC.

a) Input contact number

b) Logic state of the input

c) Contact operated time in milliseconds

Similarly, analogue value printing can also be checked by connecting the input (CON-9) to either ground or 3.3V momentarily and observing the printed results with timing. Also, the delay provided for analogue value recording may be changed to the user's requirement, as this delay will affect the recording interval of digital inputs as well. The serial monitor printed results can be copied and saved to a text file for further analysis and record.

The fault finder provides fairly accurate timing to the resolution of 1 millisecond. The uncertainty of the recorded time can be a maximum of 10 milliseconds, which is the bouncing and settling time of mechanical input contacts.

Caution. 1. Only potential-free external contacts should be used as input contacts (CON1 through CON8) with a common +5V. Otherwise, external voltage injected into the circuit will damage the STM32 board.

2. If long cables are used for input, EMI suppression circuits/opto-coupler circuits should be added to prevent spurious operation or damage.

EFY note. This device can also be constructed with Arduino boards by suitably modifying the program for digital and analogue input pins definition, and analogue value calculation. The STM32 board has been chosen for its higher clock speed. EFY

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